AI QUIZ

Backend API Specification

AI-Quiz-Ruby

REST Endpoints • WebSocket Channels • Authentication

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Overview

This document details all API endpoints and WebSocket channels available in the AI Quiz Ruby Backend (AI-Quiz-Ruby) as of the latest version of develop. It covers endpoint sections, HTTP methods, route paths, required parameters/headers, authentication mechanisms, response payloads, and real-time ActionCable events.

Authentication & Headers

Protected endpoints require a Bearer Token sent in the HTTP Authorization header.

Host Authentication (authenticate_host!)

Used for host operations such as creating games (POST /hosts/:id/create_game), advancing game state (POST /games/:id/advance_step), and creating questions (POST /questions).

Header Name	Authorization
Format	Bearer <HOST_TOKEN>

Unauthorized Response (401)

```
{
  "error": "You are not authorized to perform this action."
}
```

Forbidden Response (403)

```
{
  "error": "You do not have permission to access this resource."
}
```

Player Authentication (authenticate_player!)

Used for player operations such as submitting answers (POST /submissions).

Header Name	Authorization
Format	Bearer <PLAYER_TOKEN>

Unauthorized Response (401)

```
{
  "error": "You are not authorized to perform this action."
}
```

1. Health Check / System

Check Server Health
`GET /up`

Section	System Health

Auth Required	No
Description	Used by load balancers or uptime monitors to check if the Rails application is live.

Success Response

Status 200 OK (HTML or empty response).

---

2. Hosts Section

List All Hosts

`GET /hosts`

Section	Host Management
Auth Required	No
Description	Returns a list of all host records in the database.

Success Response

```
[
  {
    "id": 1,
    "name": "Quiz Master Alex",
    "token": "4f9d0e1a-8c2b-4e12-b13c-7a9e0f31c25a",
    "created_at": "2026-09-10T10:00:00.000Z",
    "updated_at": "2026-09-10T10:00:00.000Z"
  }
]
```

Register / Create Host

`POST /hosts`

Section	Host Management & Auth
Auth Required	No
Description	Registers a new host and generates a unique authentication token (token).

Request Body

```
{
  "host": {
    "name": "Quiz Master Alex"
  }
}
```

Success Response

```
{
  "id": 1,
  "token": "4f9d0e1a-8c2b-4e12-b13c-7a9e0f31c25a"
}
```

Host Creates a Game
`POST /hosts/:id/create_game`

Section	Host Management / Game Creation
Auth Required	Yes (Bearer <HOST_TOKEN>)
Description	Authenticated host creates a new game session associated with their host ID. Automatically populates game quiz data/rounds, generates a unique 6-character uppercase hex game_code, and broadcasts game_created and host_joined events via ActionCable to stream game_room_<game_code>.
Path Parameters	id — Host ID

Success Response

```
{
  "id": 10,
  "code": "A3F8B2"
}
```

---

3. Games Section

List All Games
`GET /games`

Section	Game Management
Auth Required	No
Description	Retrieves all games in the system.

Success Response

```
[
  {
    "id": 10,
    "game_code": "A3F8B2",
    "host_id": 1,
    "player_limit": 8,
    "status": "waiting",
    "current_round_id": null,
    "current_question_id": null,
    "question_started_at": null,
    "created_at": "2026-09-10T10:05:00.000Z",
    "updated_at": "2026-09-10T10:05:00.000Z"
  }
]
```

Join Game / Create Player
`POST /games/create_player`

Section	Game Management / Player Lobby
Auth Required	No

Description Joins a new player to the game matching game_code. Checks player limit (default limit is 8). On success, generates a player token and broadcasts a player_joined event via ActionCable.

Request Body

```
{
  "game_code": "A3F8B2",
  "player": {
    "username": "PlayerOne"
  }
}
```

Success Response

```
{
  "id": 5,
  "username": "PlayerOne",
  "score": 0,
  "eliminated": false,
  "game_id": 10,
  "token": "7b8a9c0d1e2f3a4b...",
  "created_at": "2026-09-10T10:06:00.000Z",
  "updated_at": "2026-09-10T10:06:00.000Z"
}
```

Error Response (400 — Missing Game Code)

```
{
  "error": "game_code is required"
}
```

Error Response (422 — Player Limit Exceeded)

```
{
  "error": "has reached the maximum number of players"
}
```

Error Response (422 — Validation Error)

```
{
  "username": [
    "is required"
  ]
}
```

Advance Game Step

POST /games/:id/advance_step

Section Game Control

Auth Required Yes (Bearer <HOST_TOKEN>)

Description Advances the game state for the host's game (starts game, advances to next question/round, or marks game as completed). Updates current_round_id, current_question_id, sets question_started_at, and updates the question's shown_at timestamp.

Path Parameters id — Game ID

Success Response

```
{
  "game_id": 10,
  "status": "in_progress",
  "current_round_id": 1,
```

```
  "current_question": {
    "id": 1,
    "title": "What is the capital of France?",
    "question_type": "multiple_choice",
    "position": 1,
    "round_id": 1,
    "shown_at": "2026-09-10T14:00:00.000Z",
    "options": [
      { "id": 1, "text": "London", "position": 1 },
      { "id": 2, "text": "Paris", "position": 2 }
    ]
  },
  "question_started_at": "2026-09-10T14:00:00.000Z"
}
```

Error Response (422 Unprocessable Entity)

```
{
  "error": "Unable to advance game state"
}
```

Get Players for a Game

`GET /games/get_players`

Section	Game Management / Lobby Info
Auth Required	No
Description	Returns all players for a specific game by game_code (returning id and username).
Query Parameters	game_code — Game Code (e.g. A3F8B2)

Success Response

```
[
  {
    "id": 5,
    "username": "PlayerOne"
  },
  {
    "id": 6,
    "username": "PlayerTwo"
  }
]
```

---

4. Players Section

List All Players

`GET /players`

Section	Player Management
Auth Required	No
Description	Lists all players across all games.

Success Response

```
[
  {
    "id": 5,
    "username": "PlayerOne",
    "score": 100,
    "eliminated": false,
    "game_id": 10,
    "token": "7b8a9c0d1e2f3a4b...",
    "created_at": "2026-09-10T10:06:00.000Z",
    "updated_at": "2026-09-10T10:06:00.000Z"
  }
]
```

Get Player Details

`GET /players/:id`

Section	Player Management
Auth Required	No
Description	Returns details for a specific player by ID.
Path Parameters	id — Player ID

Success Response

```
{
  "id": 5,
  "username": "PlayerOne",
  "score": 100,
  "eliminated": false,
  "game_id": 10,
  "token": "7b8a9c0d1e2f3a4b...",
  "created_at": "2026-09-10T10:06:00.000Z",
  "updated_at": "2026-09-10T10:06:00.000Z"
}
```

5. Submissions Section

Submit Answer

`POST /submissions`

Section	Gameplay / Submissions
Auth Required	Yes (Bearer <PLAYER_TOKEN>)
Description	Submits an answer for the specified question. Supports all question types. Calculates response time based on game.question_started_at, checks answer correctness according to the question type, updates the player's total score (ScoreCalculator & ScoreUpdater), and records the submission.

Question Types & Expected Answer Formats

- **true_false / single_choice / multiple_choice —** answer is an integer or numeric string representing option position/ID. Example: "answer": 2
- **drag_drop_order —** answer is an array of integers or numeric strings representing ordered option IDs/positions. Example: "answer": [2, 1, 3]
- **type_the_answer —** answer is a string with the player's typed answer text, evaluated case-insensitively (ignoring leading/trailing whitespace) against matching option texts. Example: "answer": "Paris"

Request Body Examples

Choice Question Submission

```
{
  "submission": {
    "question_id": 1,
    "answer": 2
  }
}
```

Drag and Drop Order Submission

```
{
  "submission": {
    "question_id": 3,
    "answer": [2, 1, 3]
  }
}
```

Type the Answer Submission

```
{
  "submission": {
    "question_id": 4,
    "answer": "Paris"
  }
}
```

Scoring Rules

- **Correct Answer —** base score is 100 points.
- **Free Time Allowance —** 5,000 ms (5 seconds). Response time is computed server-side from game.question_started_at.
- **Time Penalty —** -1 point per second elapsed after 5 seconds.
- **Minimum Score for Correct Answer —** 75 points.
- **Incorrect Answer —** 0 points.

Success Response (201 Created)

```
{
  "submission": {
    "id": 1,
    "player_id": 5,
    "question_id": 1,
    "round_id": 1,
    "answer": 2,
    "response_time_ms": 3200,
    "points_awarded": 100,
    "created_at": "2026-09-10T14:05:00.000Z",
    "updated_at": "2026-09-10T14:05:00.000Z"
  },
  "correct": true,
  "points_awarded": 100
}
```

Error Response (401 Unauthorized)

```
{
  "error": "You are not authorized to perform this action."
}
```

6. Rounds Section

List All Rounds

`GET /rounds`

Section	Quiz / Round Management
Auth Required	No
Description	Retrieves all rounds, including nested questions and options. correct_answer_ids are omitted from questions in response output.

Success Response

```
[
  {
    "id": 1,
    "title": "General Knowledge",
    "game_id": 10,
    "created_at": "2026-09-10T10:10:00.000Z",
    "updated_at": "2026-09-10T10:10:00.000Z",
    "questions": [
      {
        "id": 1,
        "title": "What is the capital of France?",
        "question_type": "multiple_choice",
        "position": 1,
        "round_id": 1,
        "created_at": "2026-09-10T10:10:00.000Z",
        "updated_at": "2026-09-10T10:10:00.000Z",
        "options": [
          { "id": 1, "text": "London", "position": 1, "question_id": 1 },
          { "id": 2, "text": "Paris", "position": 2, "question_id": 1 }
        ]
      }
    ]
  }
]
```

Get Round Details

`GET /rounds/:id`

Section	Quiz / Round Management
Auth Required	No
Description	Retrieves details for a specific round including nested questions and options (omitting correct_answer_ids).

Path Parameters id — Round ID

Success Response

```
{
  "id": 1,
  "title": "General Knowledge",
  "game_id": 10,
  "created_at": "2026-09-10T10:10:00.000Z",
  "updated_at": "2026-09-10T10:10:00.000Z",
  "questions": [
    {
      "id": 1,
      "title": "What is the capital of France?",
      "question_type": "multiple_choice",
      "position": 1,
      "round_id": 1,
      "options": [
        { "id": 1, "text": "London", "position": 1 },
        { "id": 2, "text": "Paris", "position": 2 }
      ]
    }
  ]
}
```

Create a Round

POST /rounds

Section Quiz / Round Management

Auth Required No

Description Creates a round with optional nested questions and options.

Request Body

```
{
  "round": {
    "title": "Round 1 - Trivia",
    "game_id": 10,
    "questions_attributes": [
      {
        "title": "What is 2 + 2?",
        "question_type": 2,
        "correct_answer_ids": [1],
        "options_attributes": [
          { "text": "4", "position": 1 },
          { "text": "5", "position": 2 }
        ]
      }
    ]
  }
}
```

Success Response

```
{
  "id": 1,
  "title": "Round 1 - Trivia",
  "game_id": 10,
  "created_at": "2026-09-10T10:10:00.000Z",
```

```
  "updated_at": "2026-09-10T10:10:00.000Z"
}
```

Error Response (422 Unprocessable Entity)

```
{
  "errors": [
    "Title can't be blank"
  ]
}
```

---

7. Questions Section

Question Types (Enum Values)

0	true_false
1	single_choice
2	multiple_choice
3	drag_drop_order
4	type_the_answer

List All Questions

`GET /questions`

Section	Question & Options Management
Auth Required	No
Description	Fetches all questions (includes options, excludes correct_answer_ids).

Success Response

```
[
  {
    "id": 1,
    "title": "What is 2 + 2?",
    "question_type": "multiple_choice",
    "position": 1,
    "round_id": 1,
    "options": [
      { "id": 1, "text": "4", "position": 1 }
    ]
  }
]
```

Get Question Details

`GET /questions/:id`

Section	Question & Options Management

Auth Required	No
Path Parameters	id — Question ID

Success Response

```
{
  "id": 1,
  "title": "What is 2 + 2?",
  "question_type": "multiple_choice",
  "position": 1,
  "round_id": 1,
  "options": [
    { "id": 1, "text": "4", "position": 1 }
  ]
}
```

Create Question

`POST /questions`

Section	Question & Options Management
Auth Required	Yes (Bearer <HOST_TOKEN>)
Description	Creates a new question within a round along with options.

Request Body

```
{
  "question": {
    "title": "Which planet is known as the Red Planet?",
    "question_type": "multiple_choice",
    "round_id": 1,
    "position": 1,
    "correct_answer_ids": [2],
    "options_attributes": [
      { "text": "Venus", "position": 1 },
      { "text": "Mars", "position": 2 }
    ]
  }
}
```

Success Response

```
{
  "id": 2,
  "title": "Which planet is known as the Red Planet?",
  "question_type": "multiple_choice",
  "position": 1,
  "round_id": 1,
  "options": [
    { "id": 1, "text": "Venus", "position": 1 },
    { "id": 2, "text": "Mars", "position": 2 }
  ]
}
```

Error Response (422 Unprocessable Entity)

```
{
  "errors": [
    "Title can't be blank"
  ]
}
```

Update Question

`PATCH / PUT  /questions/:id`

Section	Question & Options Management
Auth Required	No
Description	Updates an existing question and its nested options. Returns question along with included options.
Path Parameters	id — Question ID

Request Body

```
{
  "question": {
    "title": "Updated Question Title",
    "options_attributes": [
      { "id": 1, "text": "New Option Text", "position": 1 }
    ]
  }
}
```

Success Response

```
{
  "id": 1,
  "title": "Updated Question Title",
  "question_type": "multiple_choice",
  "position": 1,
  "round_id": 1,
  "correct_answer_ids": [1],
  "options": [
    {
      "id": 1,
      "text": "New Option Text",
      "position": 1,
      "question_id": 1
    }
  ]
}
```

Error Response (422 Unprocessable Entity)

```
{
  "errors": [
    "Title can't be blank"
  ]
}
```

8. Real-Time WebSockets (ActionCable)

Connection URL & Authentication

WebSocket Path	/cable (e.g. ws://localhost:3000/cable)

Token passed via query parameter ?token=<TOKEN> or HTTP header token: <TOKEN>.

If the token matches a valid Host, the socket connects as current_host. If it matches a valid Player, it connects as current_player. Unauthenticated connections are rejected.

Channel: GameChannel

Subscription Params game_code — the unique code of the game (e.g., "A3F8B2")

Stream Name game_room_<game_code>

Subscription authorization: a current_player is authorized only if they belong to the game; a current_host is authorized only if they own (host_id) the game.

Incoming Socket Action: submit_answer

Client sends the submit_answer command to GameChannel.

Payload

```
{
  "command": "message",
  "identifier": "{\"channel\":\"GameChannel\",\"game_code\":\"A3F8B2\"}",
  "data": "{\"action\":\"submit_answer\",\"player_id\":5}"
}
```

Broadcast Sent to Stream

```
{
  "event": "answer_submitted",
  "player_id": 5,
  "status": "waiting_for_results"
}
```

Server Broadcast Events

`game_created`

Triggered when host calls POST /hosts/:id/create_game

```
{
  "event": "game_created",
  "game_code": "A3F8B2",
  "total_players": 0
}
```

`host_joined`

Triggered when host calls POST /hosts/:id/create_game

```
{
  "event": "host_joined",
  "host": {
    "id": 1,
    "name": "Quiz Master Alex"
  }
}
```

`player_joined`

Triggered when player calls POST /games/create_player

```
{
  "event": "player_joined",
  "player": {
    "id": 5,
    "username": "PlayerOne"
  },
```

```
  "total_players": 1,
  "player_limit": 8
}
```

answer_submitted

Triggered via the submit_answer socket action

```
{
  "event": "answer_submitted",
  "player_id": 5,
  "status": "waiting_for_results"
}
```

players_eliminated

Triggered via PlayerEliminationService after round end

```
{
  "event": "players_eliminated",
  "round_id": 1,
  "round_title": "Round 1 - General Knowledge",
  "advancing_player_ids": [5, 6],
  "eliminated_players": [
    {
      "id": 7,
      "username": "PlayerThree",
      "final_score": 0
    }
  ]
}
```